



Chapter 4

Digital Transmission

4-1 DIGITAL-TO-DIGITAL CONVERSION

*In this section, we see how we can represent digital data by using digital signals. The conversion involves three techniques: **Line Coding**, block coding, and scrambling. Line coding is always needed; block coding and scrambling may or may not be needed.*

Topics discussed in this section:

Line Coding

Line Coding Schemes

Figure 4.1 *Line coding and decoding*

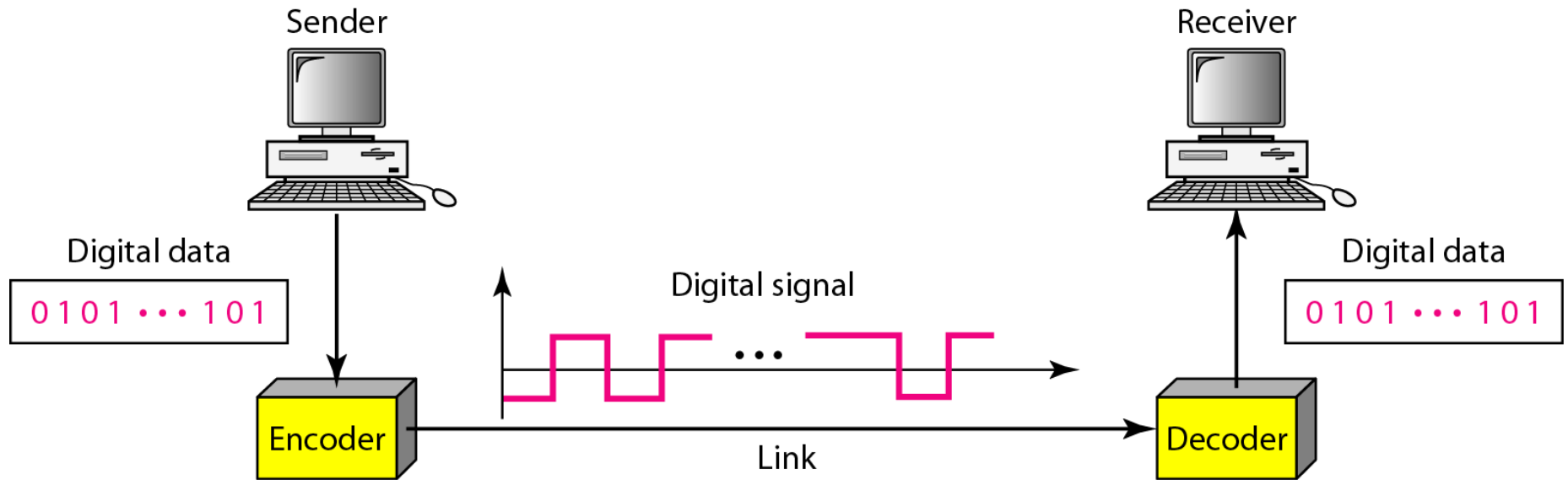
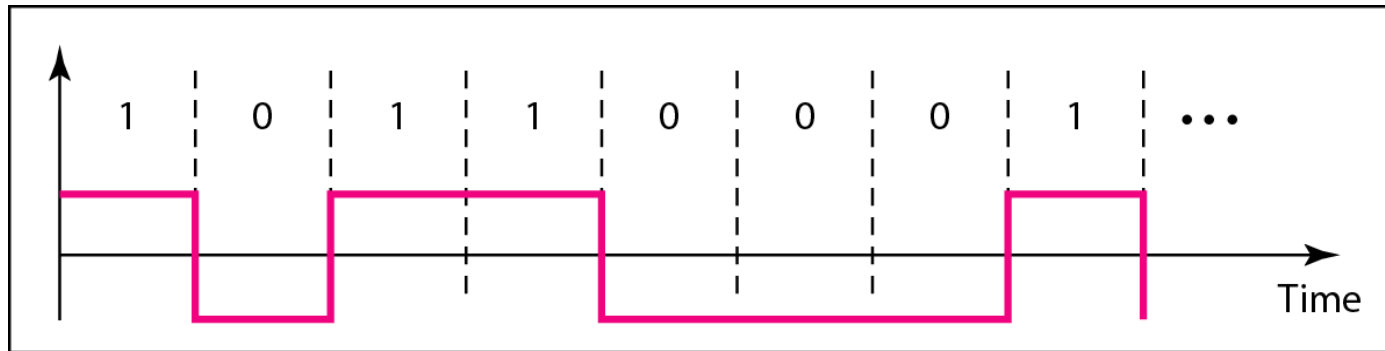
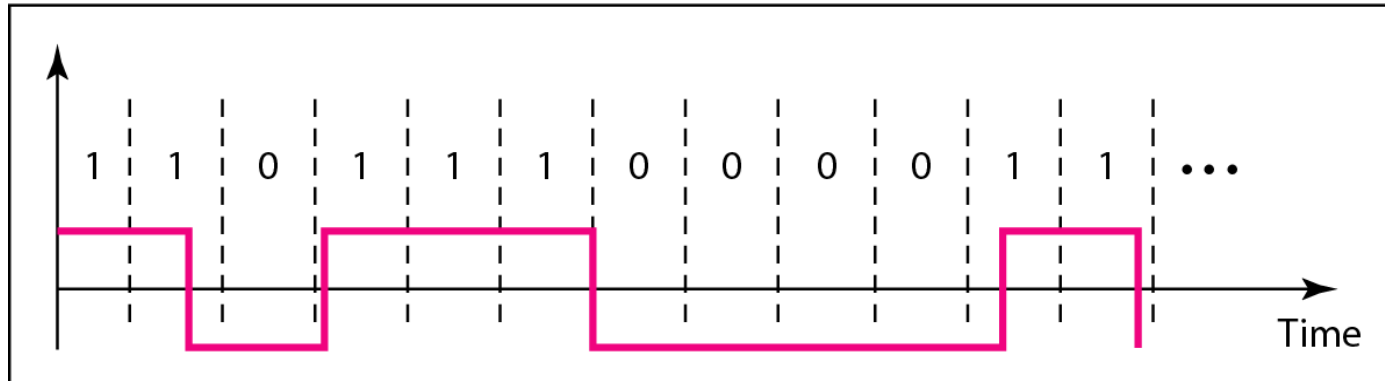


Figure 4.3 *Effect of lack of synchronization*

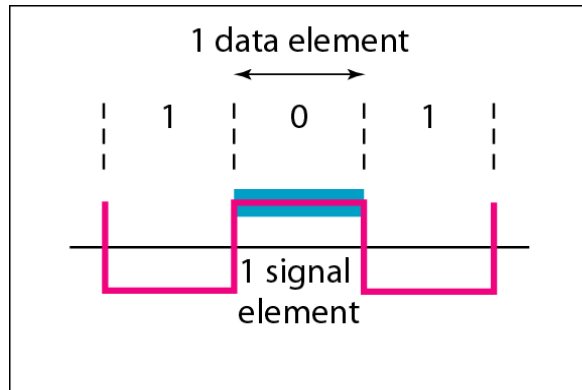


a. Sent

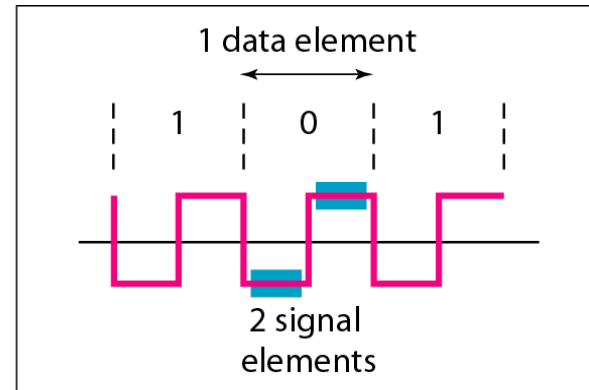


b. Received

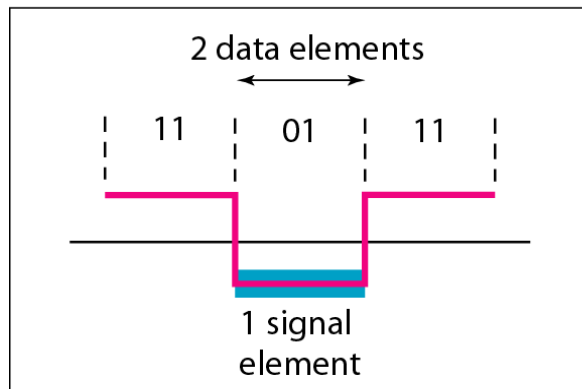
Figure 4.2 *Signal element versus data element*



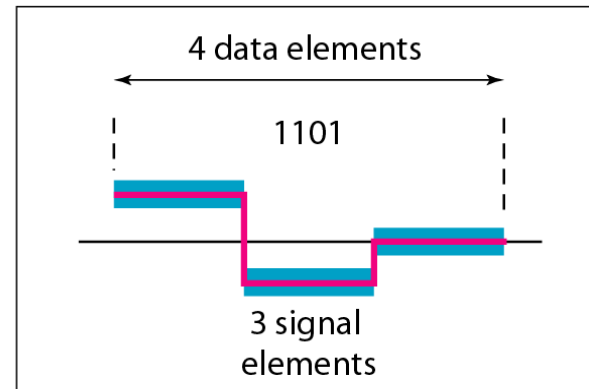
a. One data element per one signal element ($r = 1$)



b. One data element per two signal elements ($r = \frac{1}{2}$)



c. Two data elements per one signal element ($r = 2$)



d. Four data elements per three signal elements ($r = \frac{4}{3}$)

Figure 4.4 *Line coding schemes*

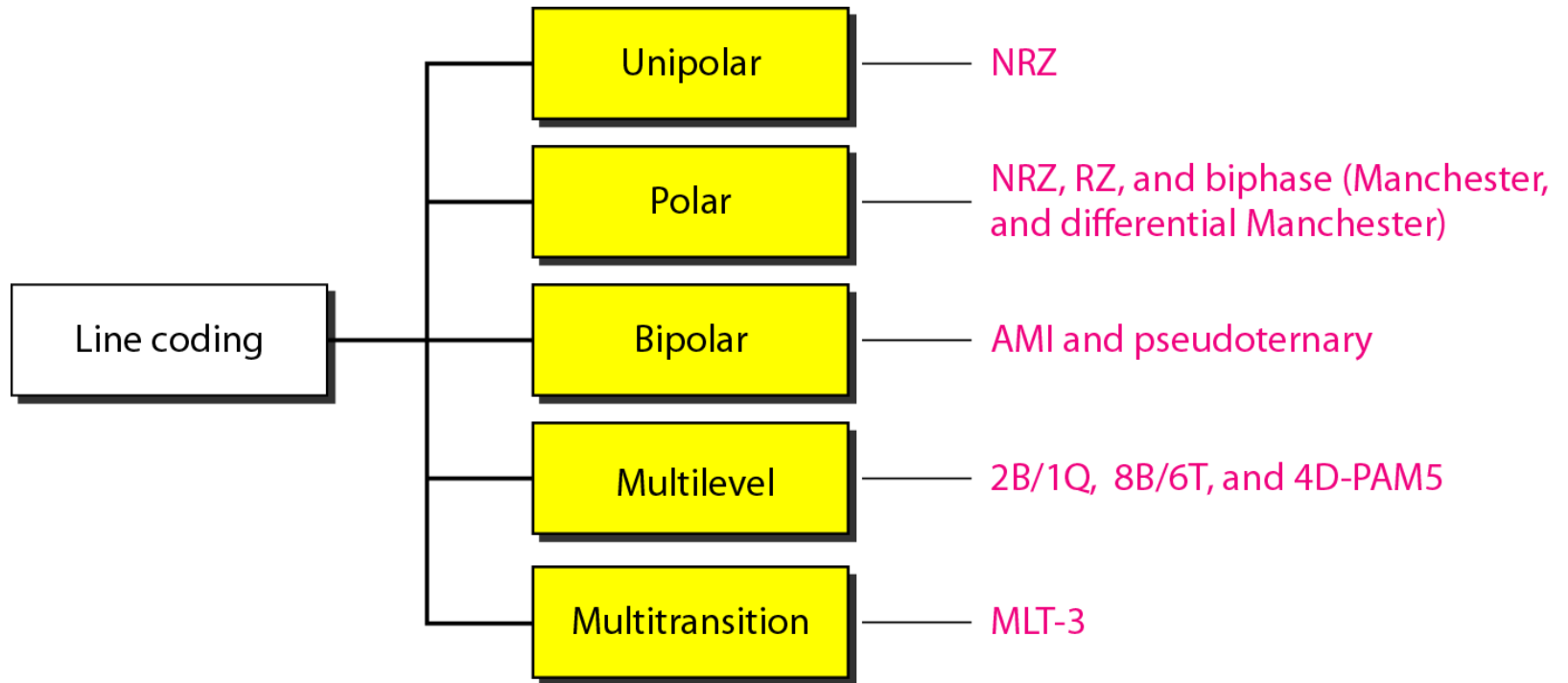
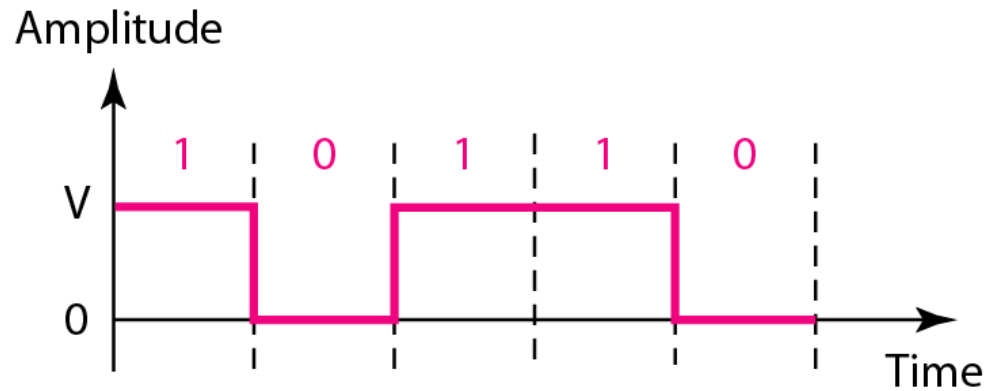


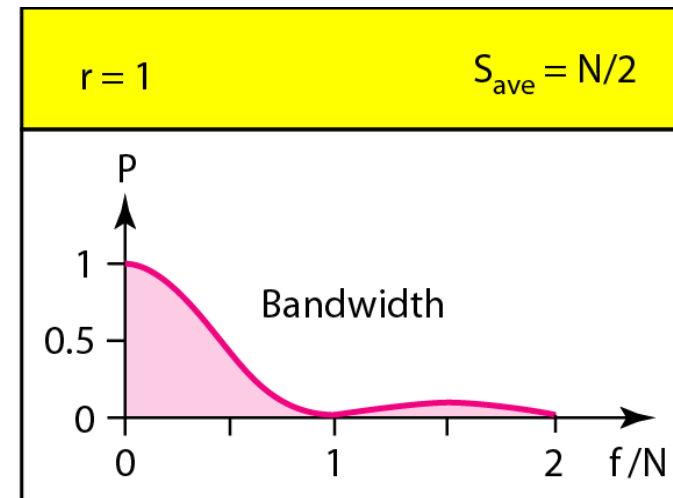
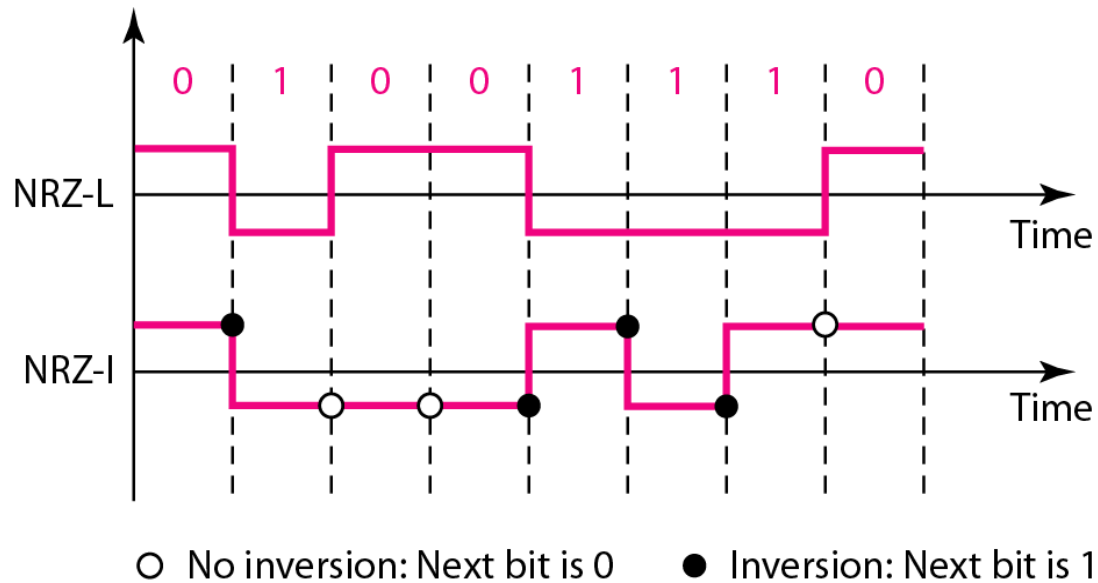
Figure 4.5 *Unipolar NRZ scheme*



$$\frac{1}{2}V^2 + \frac{1}{2}(0)^2 = \frac{1}{2}V^2$$

Normalized power

Figure 4.6 *Polar NRZ-L and NRZ-I schemes*



Note

**In NRZ-L the level of the voltage
determines the value of the bit.**

**In NRZ-I the inversion
or the lack of inversion
determines the value of the bit.**

Note

NRZ-L and NRZ-I both have an average signal rate of $N/2$ Bd.

NRZ-L and NRZ-I both have a DC component problem.

Figure 4.7 *Polar RZ scheme*

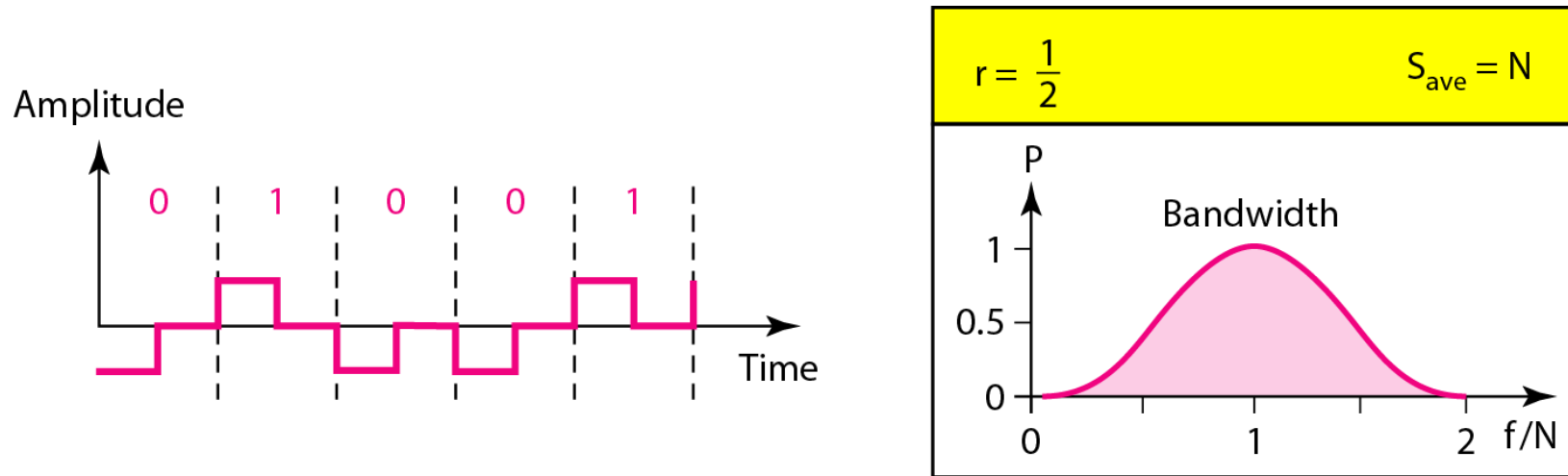
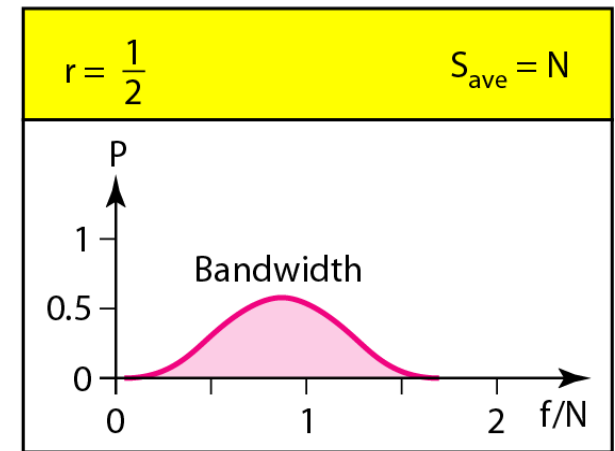
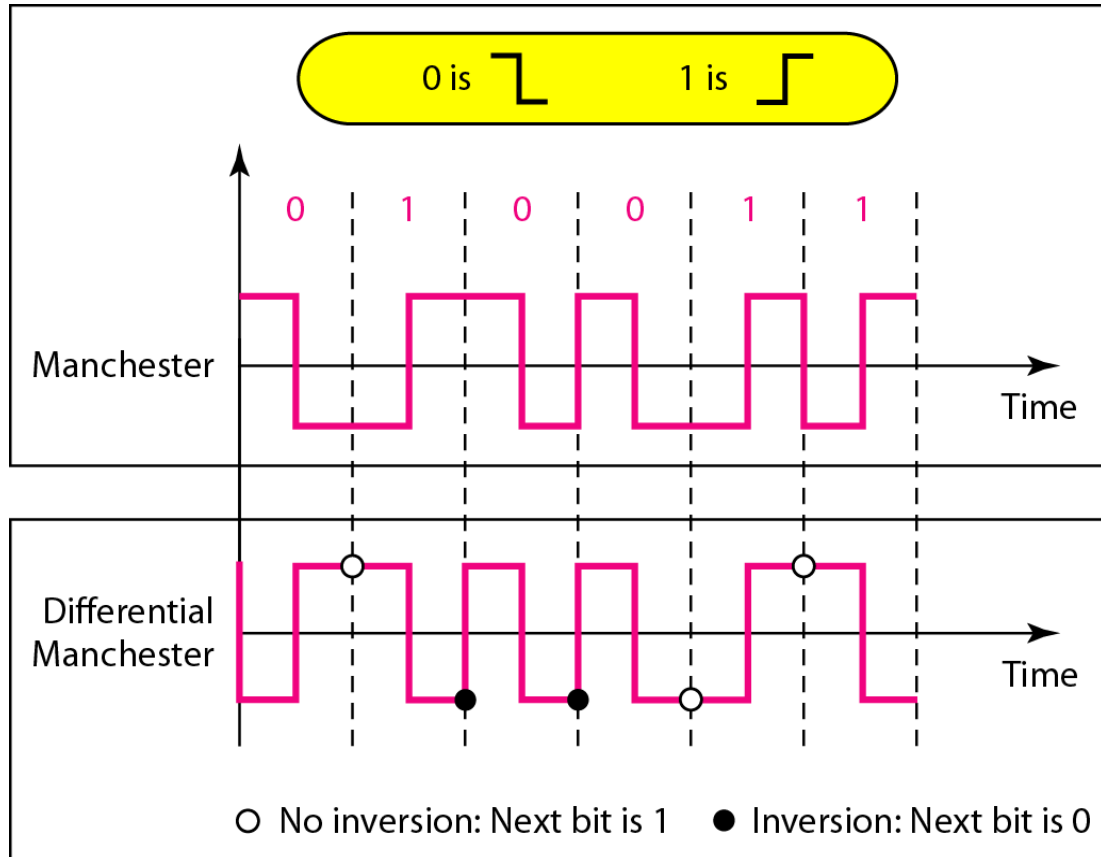


Figure 4.8 *Polar biphas: Manchester and differential Manchester schemes*

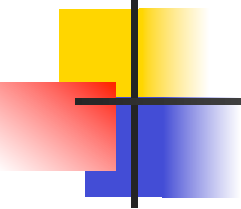


Note

In Manchester and differential Manchester encoding, the transition at the middle of the bit is used for synchronization.

Note

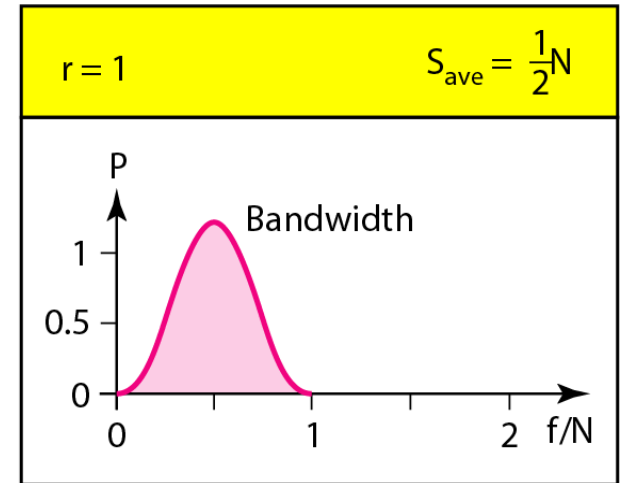
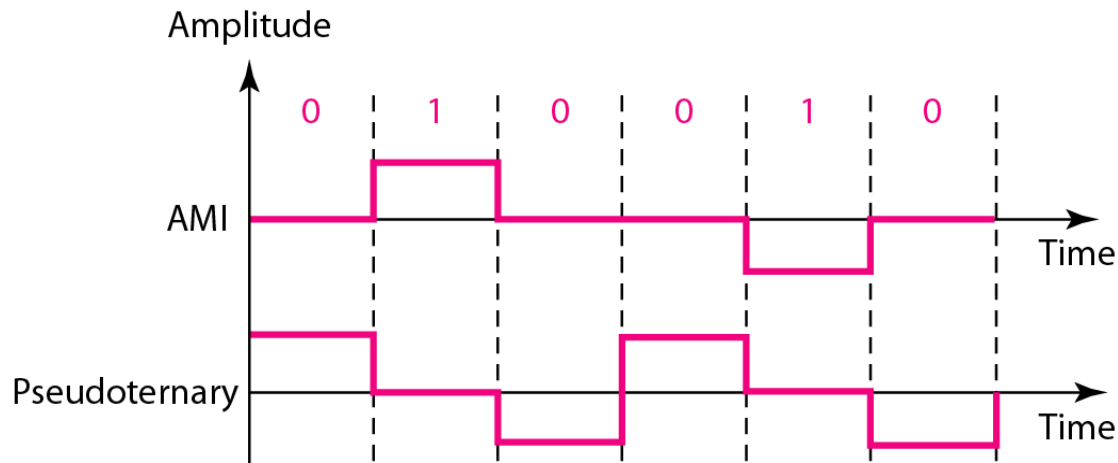
The minimum bandwidth of Manchester and differential Manchester is 2 times that of NRZ.



Note

**In bipolar encoding, we use three levels:
positive, zero, and negative.**

Figure 4.9 *Bipolar schemes: AMI and pseudoternary*



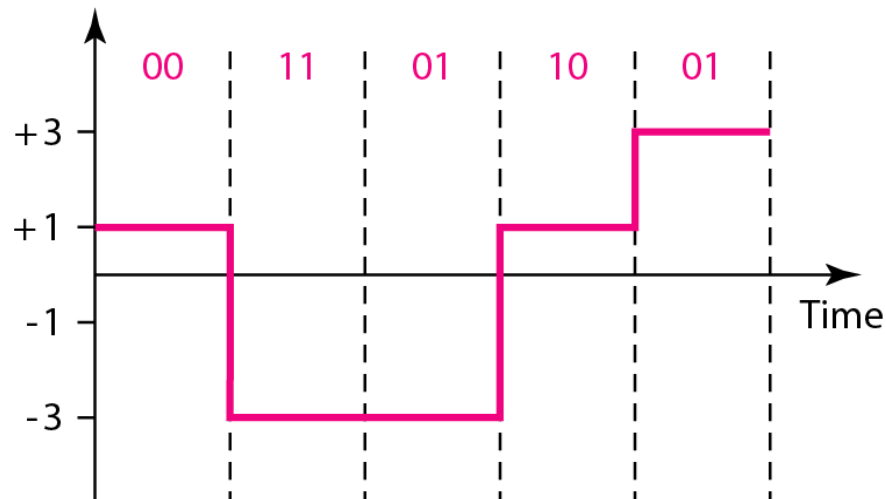
Note

In $mBnL$ schemes, a pattern of m data elements is encoded as a pattern of n signal elements in which $2^m \leq L^n$.

Figure 4.10 *Multilevel: 2B1Q scheme*

	Previous level: positive	Previous level: negative
Next bits	Next level	Next level
00	+1	-1
01	+3	-3
10	-1	+1
11	-3	+3

Transition table



Assuming positive original level

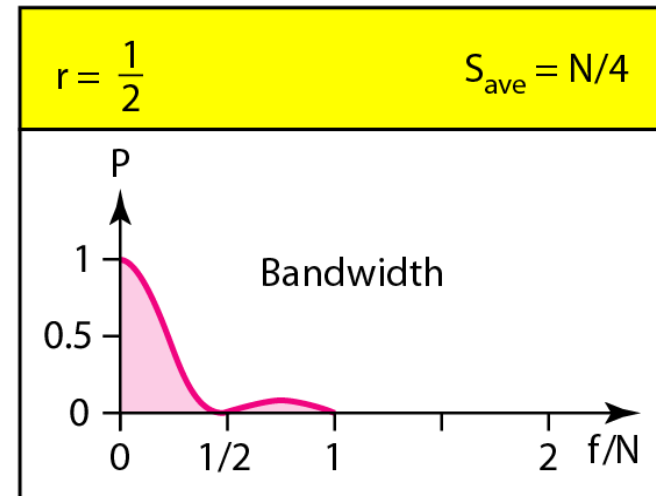


Table 4.1 *Summary of line coding schemes*

<i>Category</i>	<i>Scheme</i>	<i>Bandwidth (average)</i>	<i>Characteristics</i>
Unipolar	NRZ	$B = N/2$	Costly, no self-synchronization if long 0s or 1s, DC
Unipolar	NRZ-L	$B = N/2$	No self-synchronization if long 0s or 1s, DC
	NRZ-I	$B = N/2$	No self-synchronization for long 0s, DC
	Biphase	$B = N$	Self-synchronization, no DC, high bandwidth
Bipolar	AMI	$B = N/2$	No self-synchronization for long 0s, DC
Multilevel	2B1Q	$B = N/4$	No self-synchronization for long same double bits
	8B6T	$B = 3N/4$	Self-synchronization, no DC
	4D-PAM5	$B = N/8$	Self-synchronization, no DC
Multiline	MLT-3	$B = N/3$	No self-synchronization for long 0s

4-2 ANALOG-TO-DIGITAL CONVERSION

*We have seen in Chapter 3 that a digital signal is superior to an analog signal. The tendency today is to change an analog signal to digital data. In this section we describe two techniques, **pulse code modulation** and **delta modulation**.*

Topics discussed in this section:

Pulse Code Modulation (PCM)

Delta Modulation (DM)

Figure 4.21 *Components of PCM encoder*

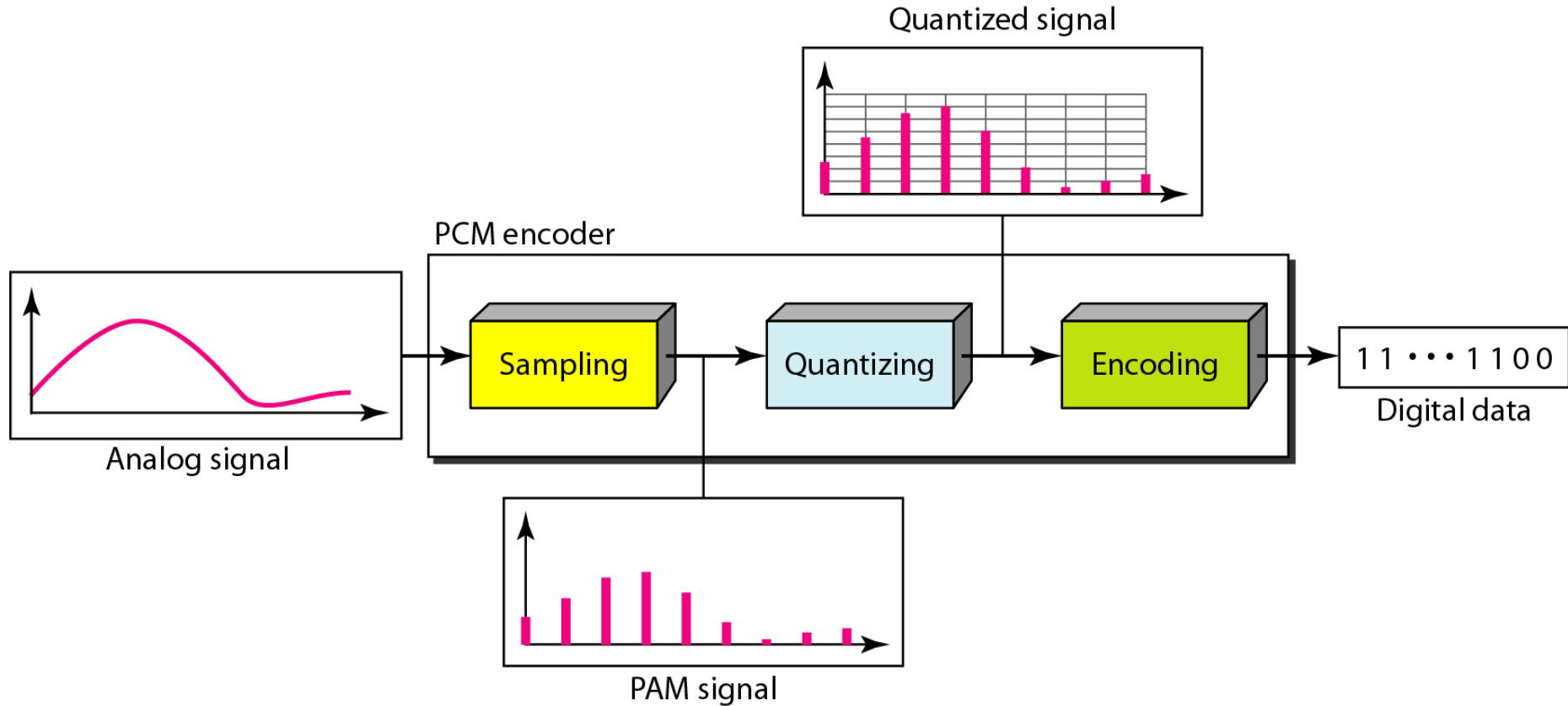
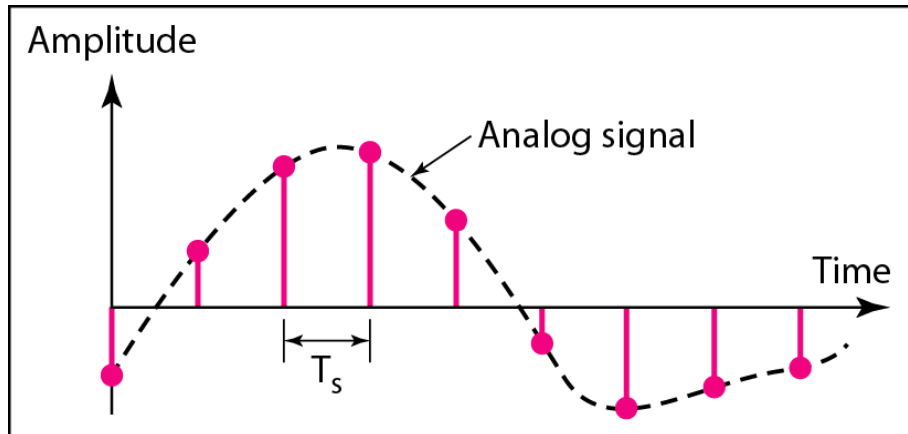
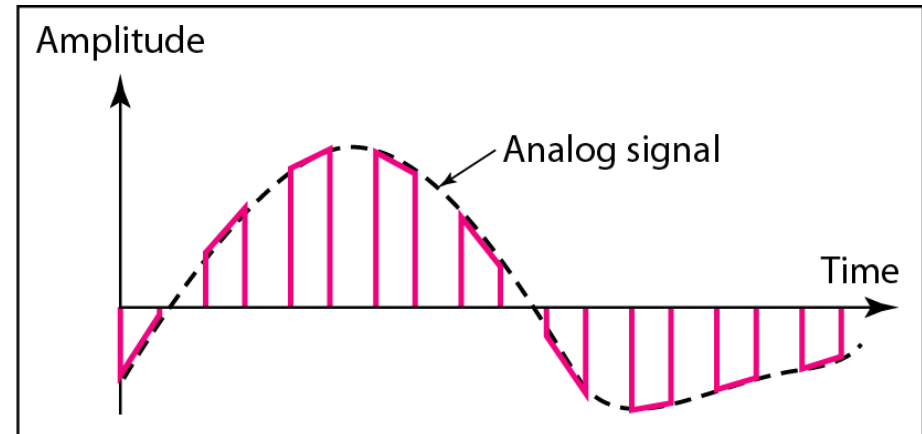


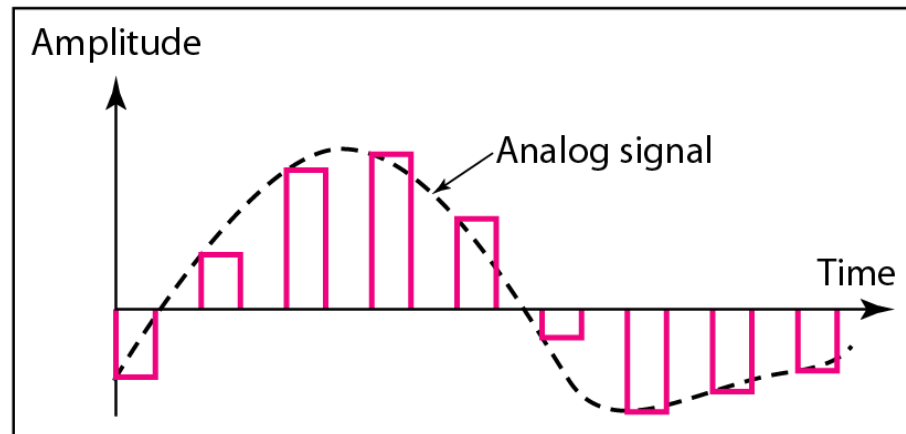
Figure 4.22 *Three different sampling methods for PCM*



a. Ideal sampling



b. Natural sampling

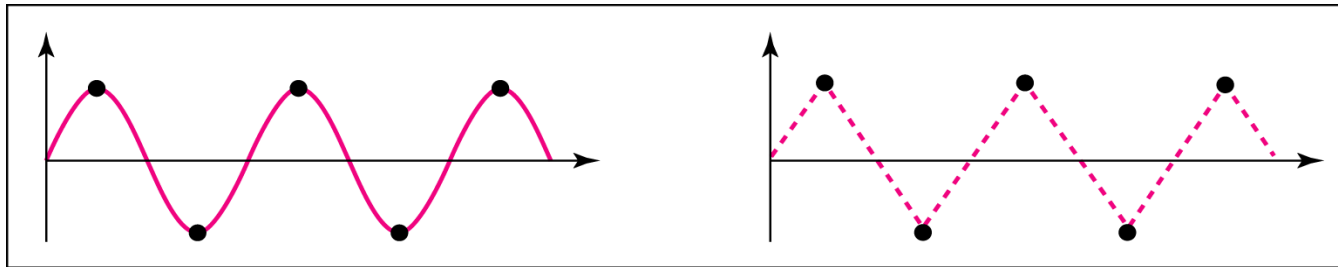


c. Flat-top sampling

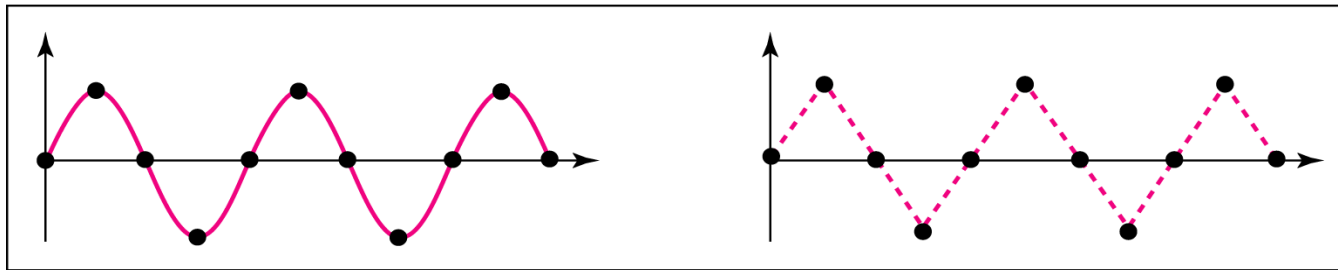
Note

According to the Nyquist theorem, the sampling rate must be at least 2 times the highest frequency contained in the signal.

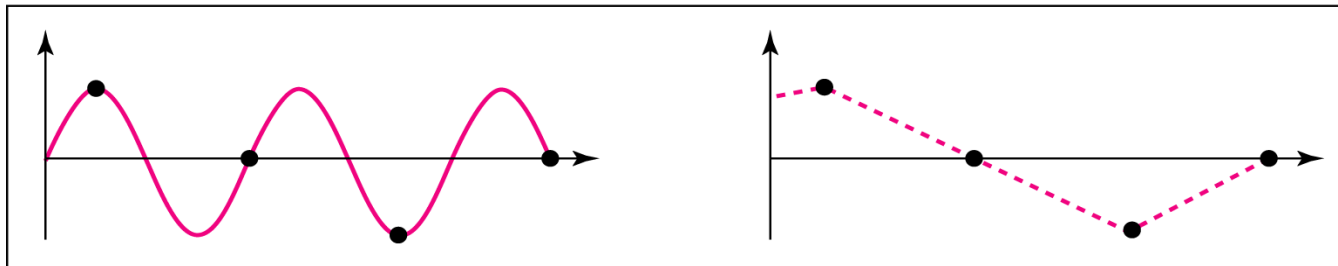
Figure 4.24 *Recovery of a sampled sine wave for different sampling rates*



a. Nyquist rate sampling: $f_s = 2 f$



b. Oversampling: $f_s = 4 f$



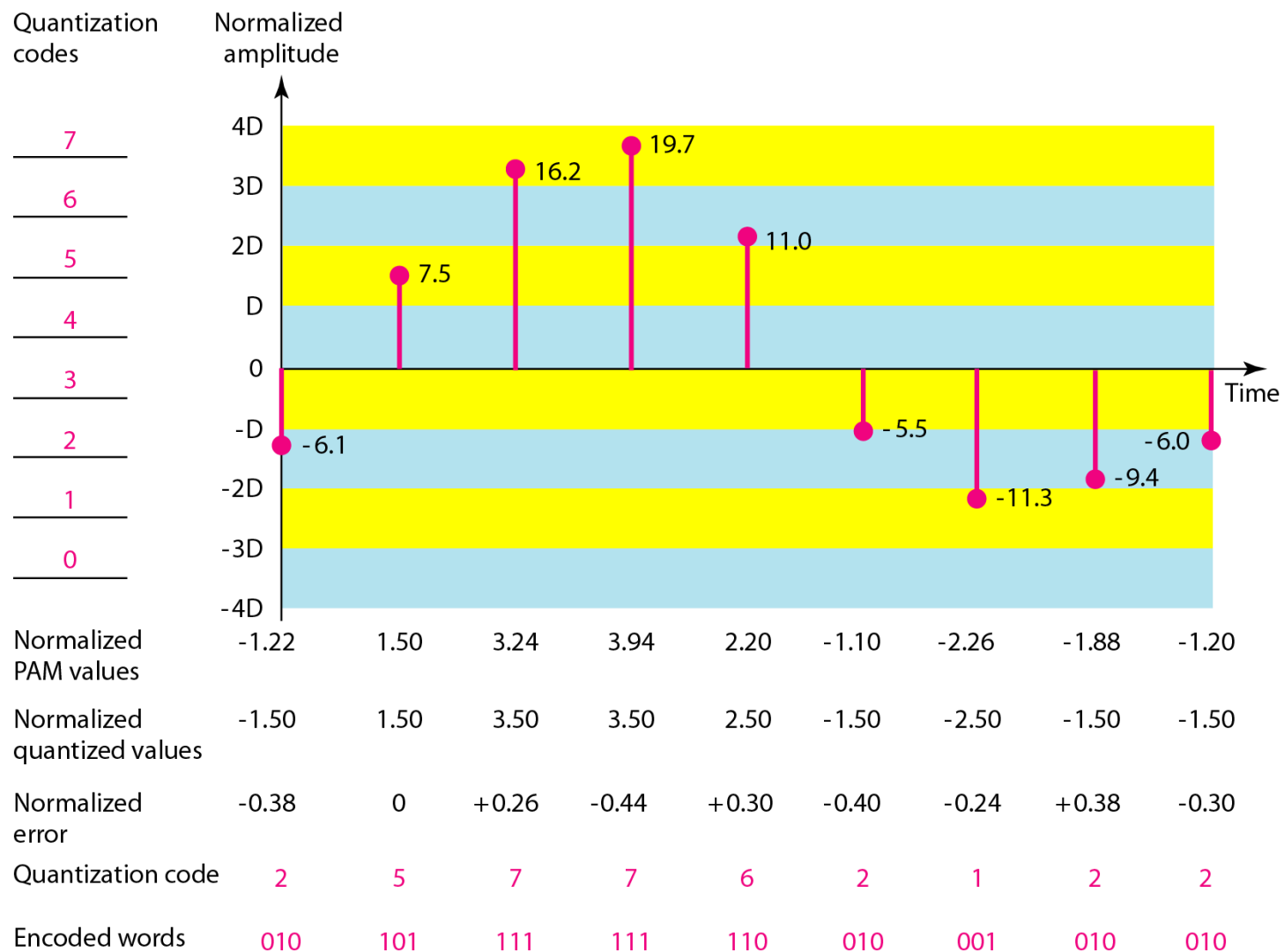
c. Undersampling: $f_s = f$



Example 4.9

Telephone companies digitize voice by assuming a maximum frequency of 4000 Hz. The sampling rate therefore is 8000 samples per second.

Figure 4.26 *Quantization and encoding of a sampled signal*





Example 4.14

We want to digitize the human voice. What is the bit rate, assuming 8 bits per sample?

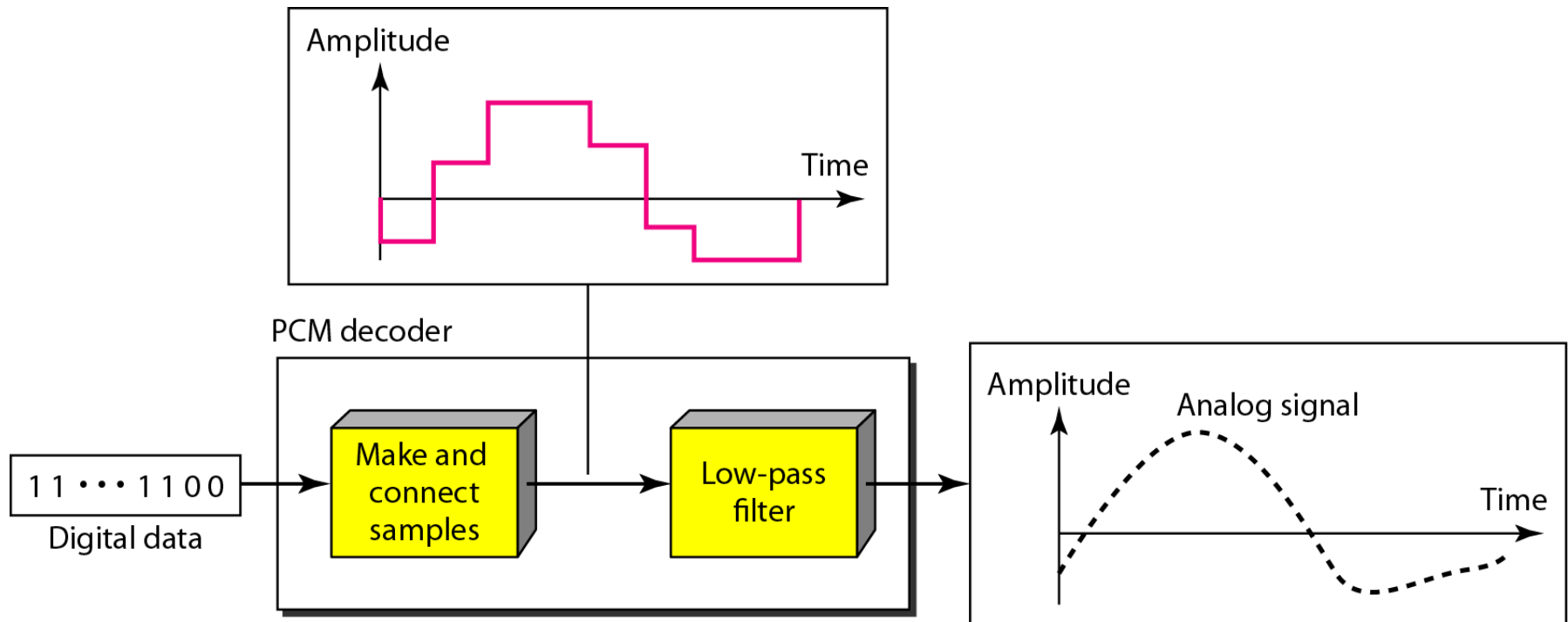
Solution

The human voice normally contains frequencies from 0 to 4000 Hz. So the sampling rate and bit rate are calculated as follows:

$$\text{Sampling rate} = 4000 \times 2 = 8000 \text{ samples/s}$$

$$\text{Bit rate} = 8000 \times 8 = 64,000 \text{ bps} = 64 \text{ kbps}$$

Figure 4.27 *Components of a PCM decoder*



4-3 TRANSMISSION MODES

The transmission of binary data across a link can be accomplished in either parallel or serial mode. In parallel mode, multiple bits are sent with each clock tick. In serial mode, 1 bit is sent with each clock tick. While there is only one way to send parallel data, there are three subclasses of serial transmission: asynchronous, synchronous, and isochronous.

Topics discussed in this section:

Parallel Transmission

Serial Transmission

Figure 4.31 *Data transmission and modes*

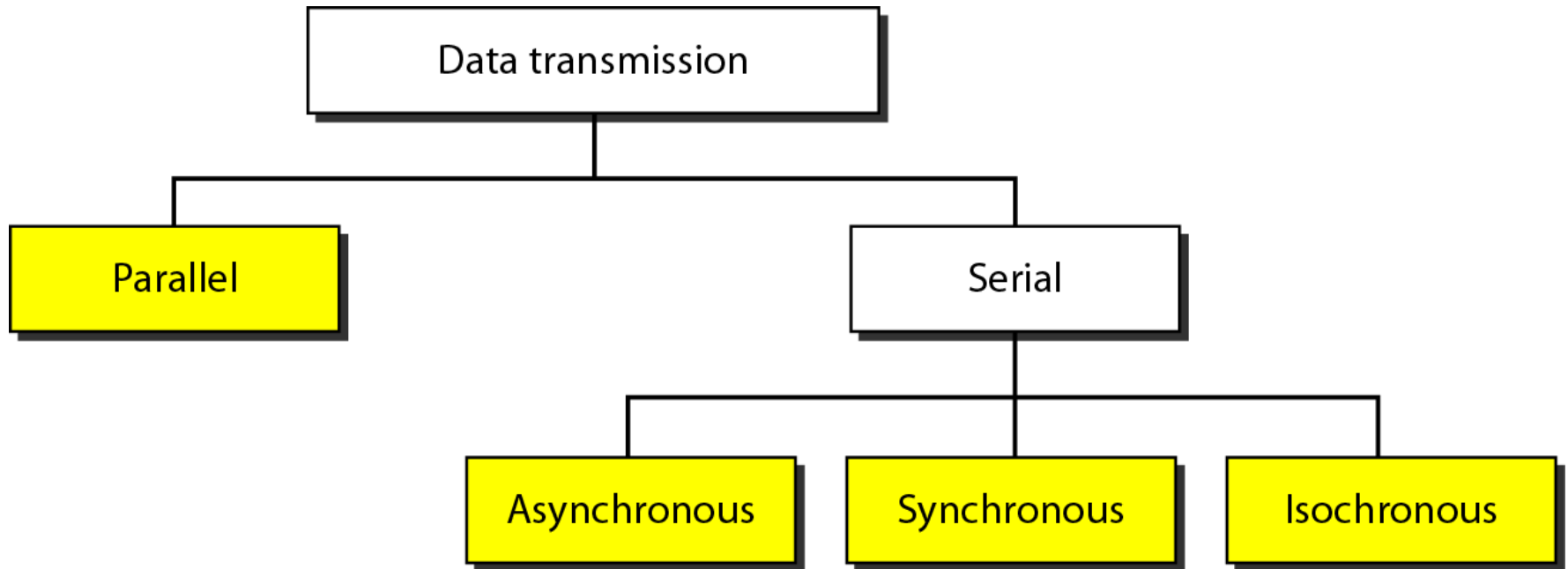


Figure 4.32 *Parallel transmission*

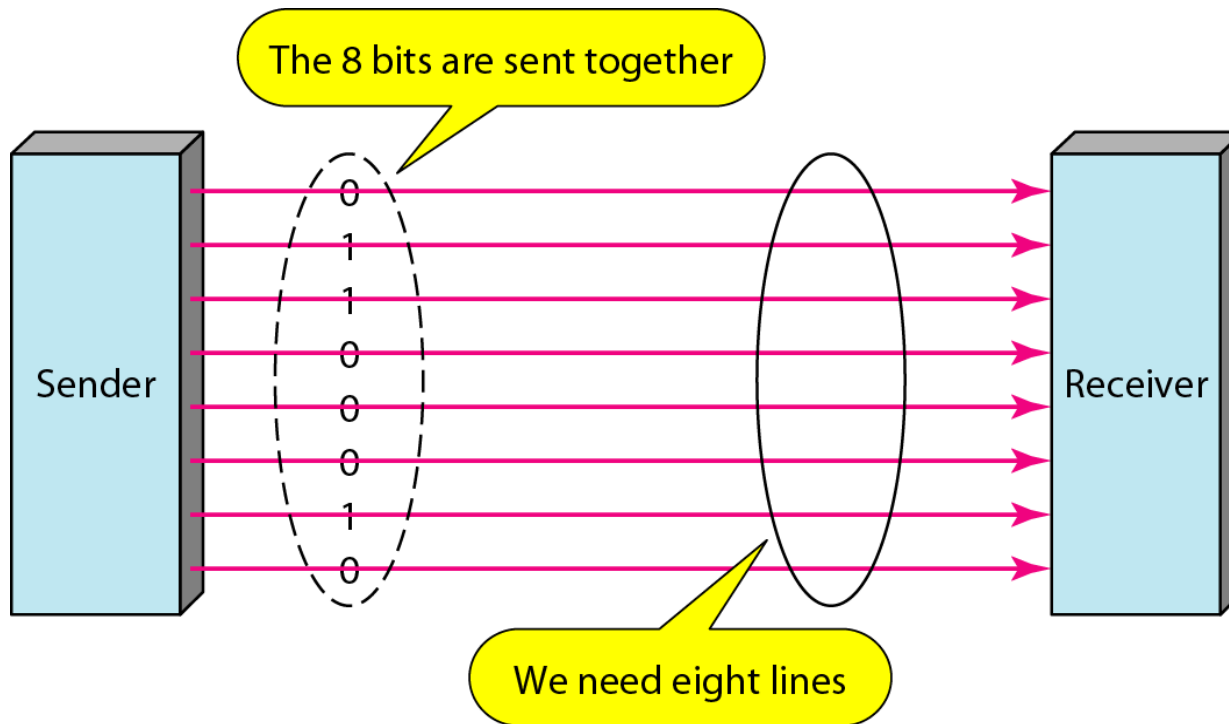
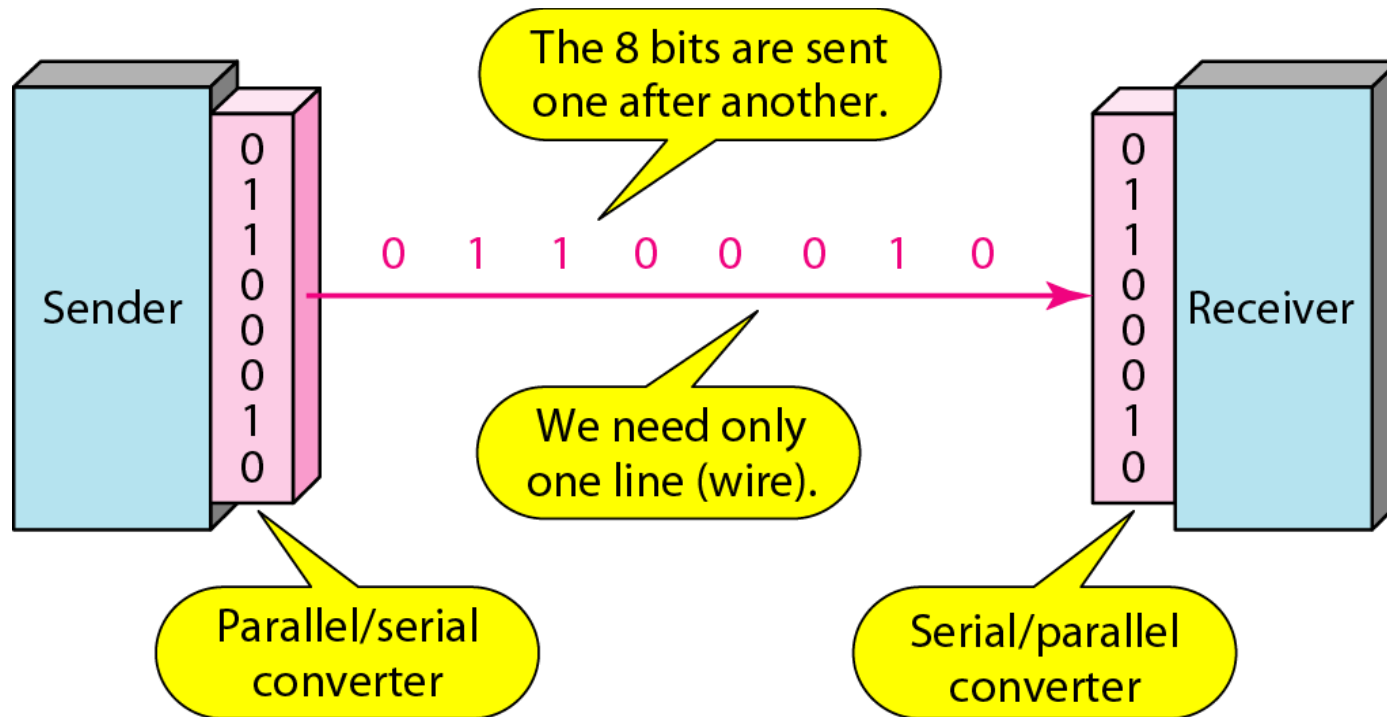


Figure 4.33 *Serial transmission*

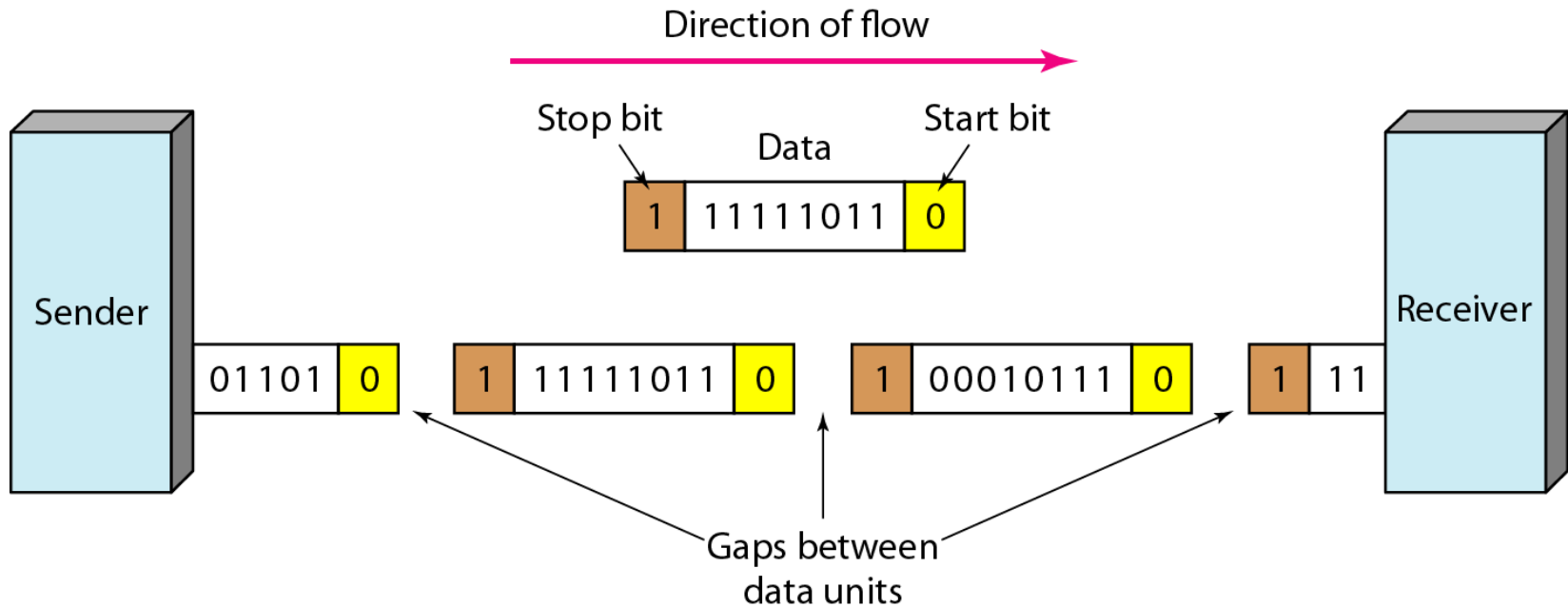


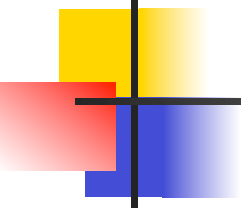
Note

In asynchronous transmission, we send 1 start bit (0) at the beginning and 1 or more stop bits (1s) at the end of each byte. There may be a gap between each byte.

Asynchronous here means “asynchronous at the byte level,” but the bits are still synchronized; their durations are the same.

Figure 4.34 *Asynchronous transmission*





Note

In synchronous transmission, we send bits one after another without start or stop bits or gaps. It is the responsibility of the receiver to group the bits.

Figure 4.35 *Synchronous transmission*

